



A Fairness Conscious Cache Replacement Policy for Last Level Cache

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MOTIVATION

Cache-friendly (Cf)
good locality

Non-cache-friendly
non-Cf: frequent misses

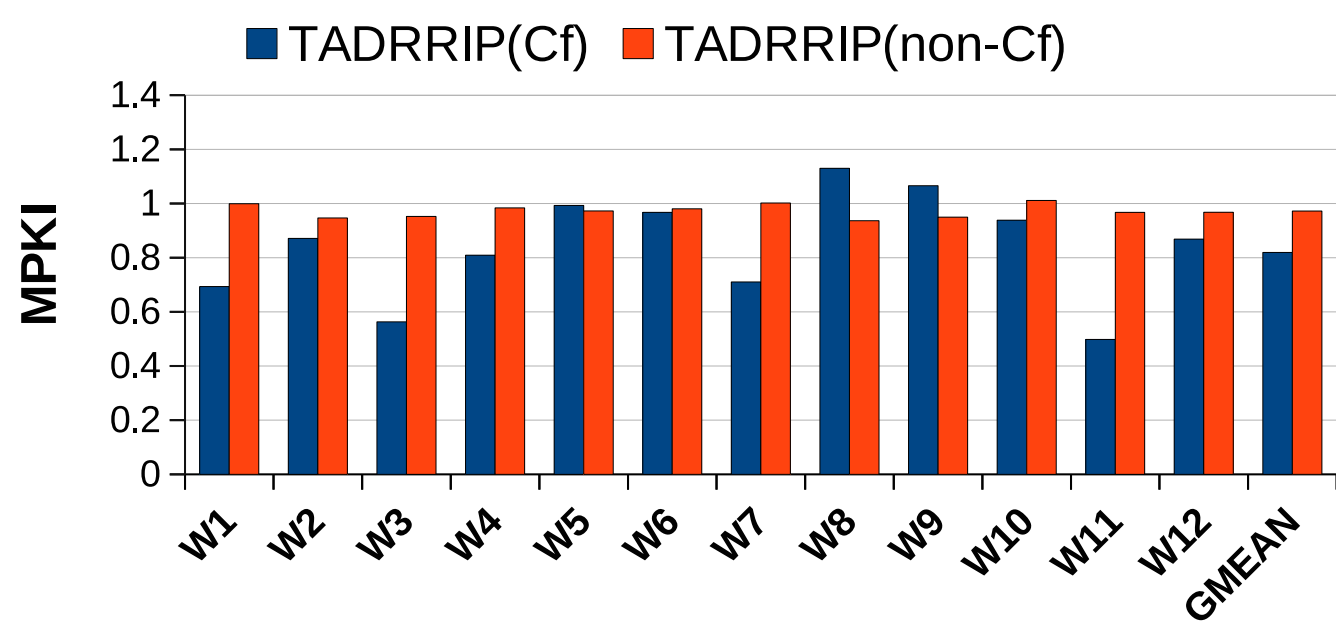
SHARED LAST LEVEL CACHE

Applications on different cores share the LLC. Non-Cf misses can **prematurely evict useful Cf blocks**, causing cache pollution, thrashing and unfair LLC usage.

How can shared LLC replacement stay fair while preserving useful blocks?

WHY EXISTING POLICIES STRUGGLE

MPKI normalized to SRRIP; lower is better.



TADRRIP panel; 2 cores, 12 mixes: MPKI falls by 18% for Cf and 2.7% for non-Cf. Existing policies miss the combined problem.

APPLICATION-AWARE EVICTION

Full set; incoming block from application X .
 $\maxRRPV = 2^M - 1$.

- An X block has maxRRPV**
Prefer that block as the victim.
- Only other owners have maxRRPV**
Choose a victim; round-robin across eligible applications if needed.
- No block has maxRRPV**
Age **only X 's blocks** in this set.

One application's misses should not age every other application's blocks.

INSERTION / PROMOTION

Event	RRPV
Cache hit	0
New block, invalid entry available	$2^M - 2$
Full set $\rightarrow$ select victim:	
Incoming block absent from EbIS	$2^M - 2$
Incoming block found in EbIS	0
Lower RRPV = sooner predicted reuse. An EbIS hit gives high insertion priority.	

EbIS: REMEMBER EVICTED BLOCKS

Evicted Block Information Storage: on-chip tag, set and application metadata, with **no cache-block data**.

A	A	B	A	B	A	B	B	A	B	B	A	B	B	B	B
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When full, maximize $totalBlocks - minBlockDistance$; remove that application's oldest entry.

Selection balances entry count per application and nearest distance to the front.

128 entries: a practical balance.

EXPERIMENTAL SETUP

Processor	$4 \times$ x86 OoO superscalar, 3 GHz
L1I / core	32 KB, 8-way, 1 cycle
L1D / core	32 KB, 8-way, 4 cycles
L2 / core	512 KB, 8-way, 10 cycles
Shared L3 / LLC	4 MB, 16-way, 20 cycles
All cache blocks	64 B
DRAM	DDR3, 4 GB, 64-bit channel

ChampSim [4] • SPEC CPU 2006 [5]

1 million warm-up + 200 million measured instructions. **20 four-core workload mixes**; one application per core.

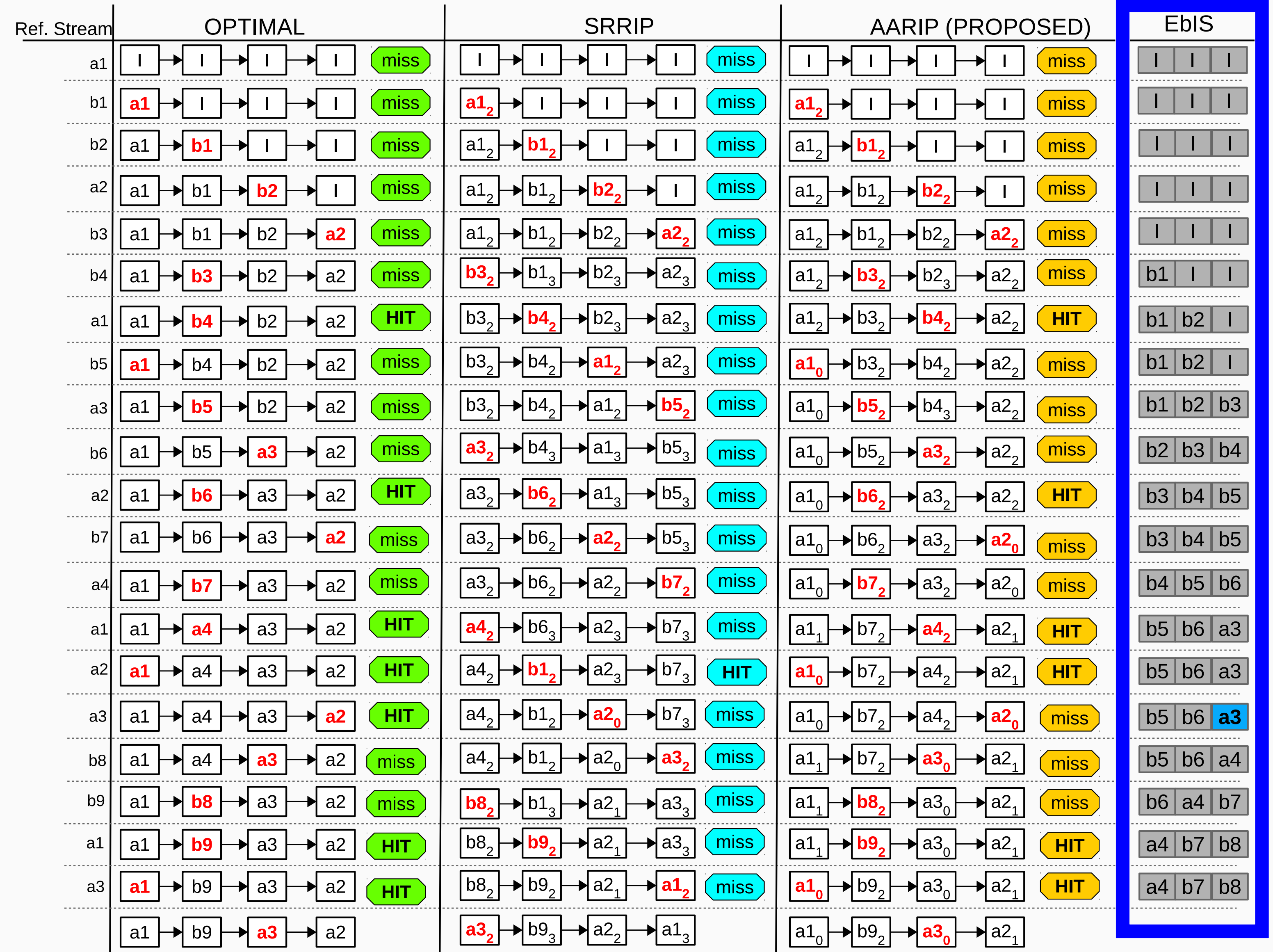
ESSENTIAL REFERENCES

- [1] RRP / TADRRIP: A. Jaleel et al. ISCA, 2010, pp. 60–71.
- [2] ABRIP: P. Lathigara, S. Balachandran, V. Singh. ICCD, 2015, pp. 172–179.
- [3] EAF: V. Seshadri et al. PACT, 2012, pp. 355–366.
- [4] *ChampSim repository*. <https://github.com/ChampSim/ChampSim>
- [5] J. L. Henning. *SPEC CPU2006 Benchmark Descriptions*. ACM SIGARCH Computer Architecture News 34(4), 2006, pp. 1–17.

AARIP Application Aware Re-reference Interval Prediction

RE-REFERENCE + APPLICATION BEHAVIOR + PREMATURE EVICTION

RRPV predicts reuse • Ownership guides victims • EbIS enables recovery



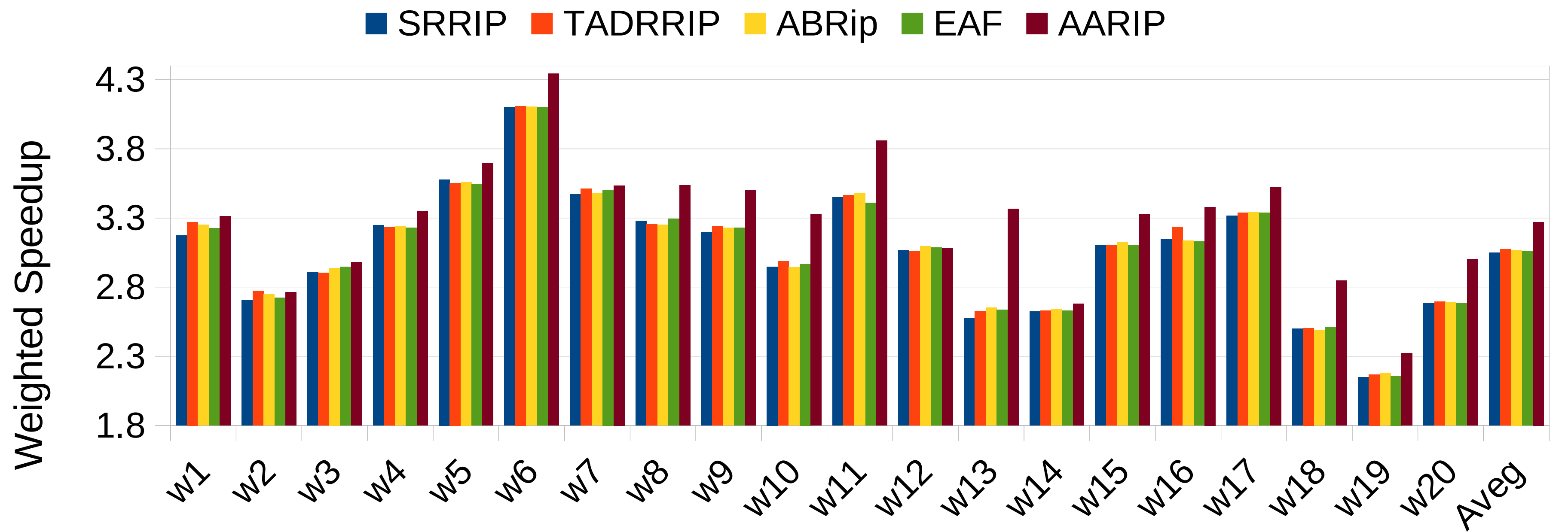
SRRIP: non-Cf B blocks displace useful Cf A blocks. **AARIP:** One 4-way set; $M = 2$; OPT is the ideal reference. AARIP operates **locally within each cache set**. application awareness protects reuse.

FOUR-CORE RESULTS Better performance, throughput and fairness

Average improvement over SRRIP • 20 workload mixes

+7.28% WEIGHTED SPEEDUP
+4.9% THROUGHPUT
+7.85% HARMONIC MEAN FAIRNESS

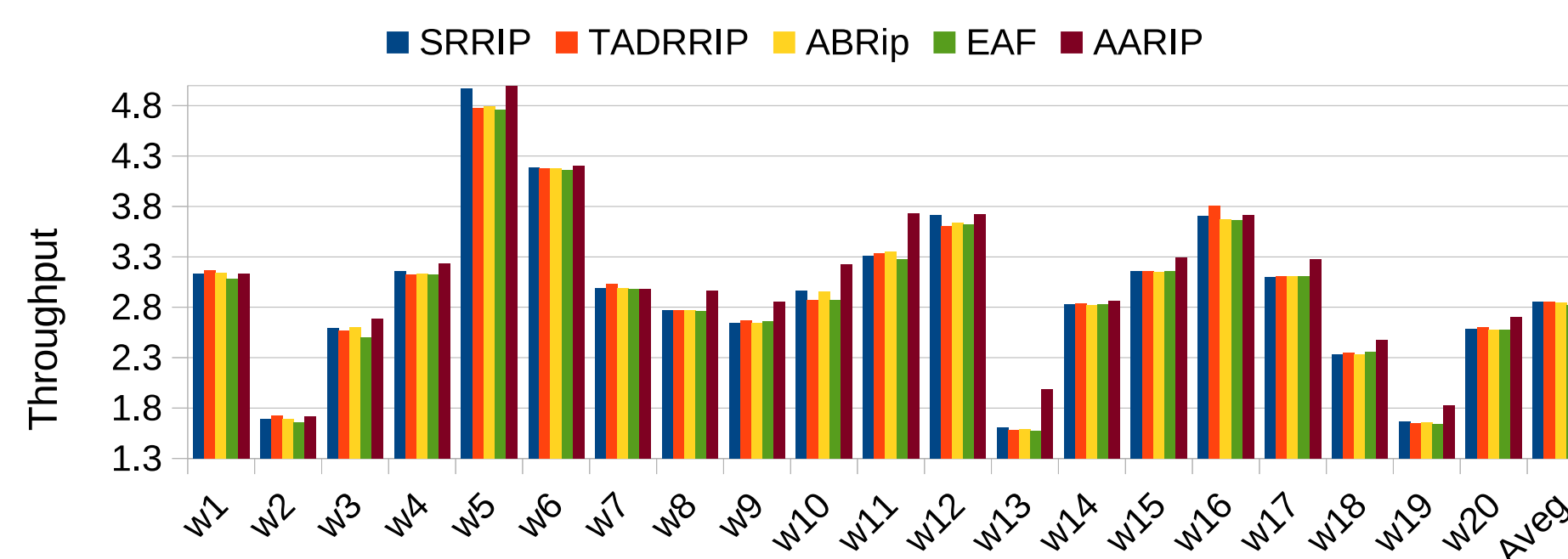
WEIGHTED SPEEDUP Overall shared-system performance



Geometric means: SRRIP 3.049 TADRRIP 3.073 ABRip 3.069 EAF 3.062 **AARIP 3.271**

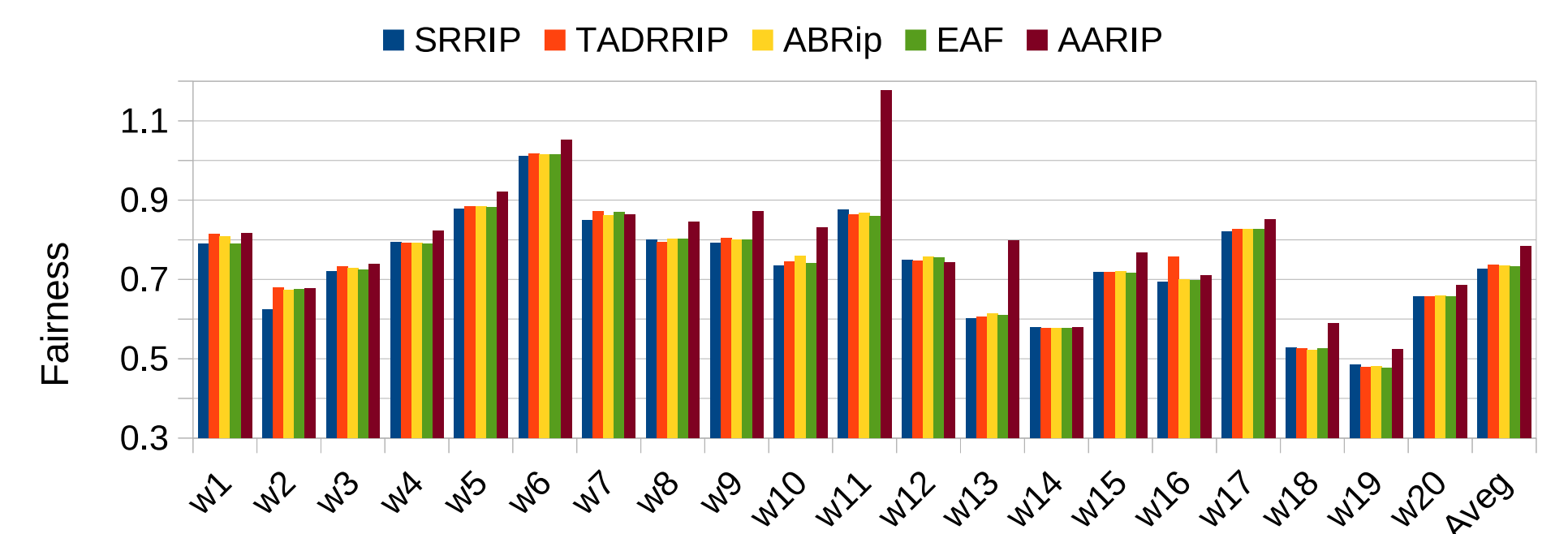
THROUGHPUT

Sum of IPC across cores



HARMONIC MEAN FAIRNESS

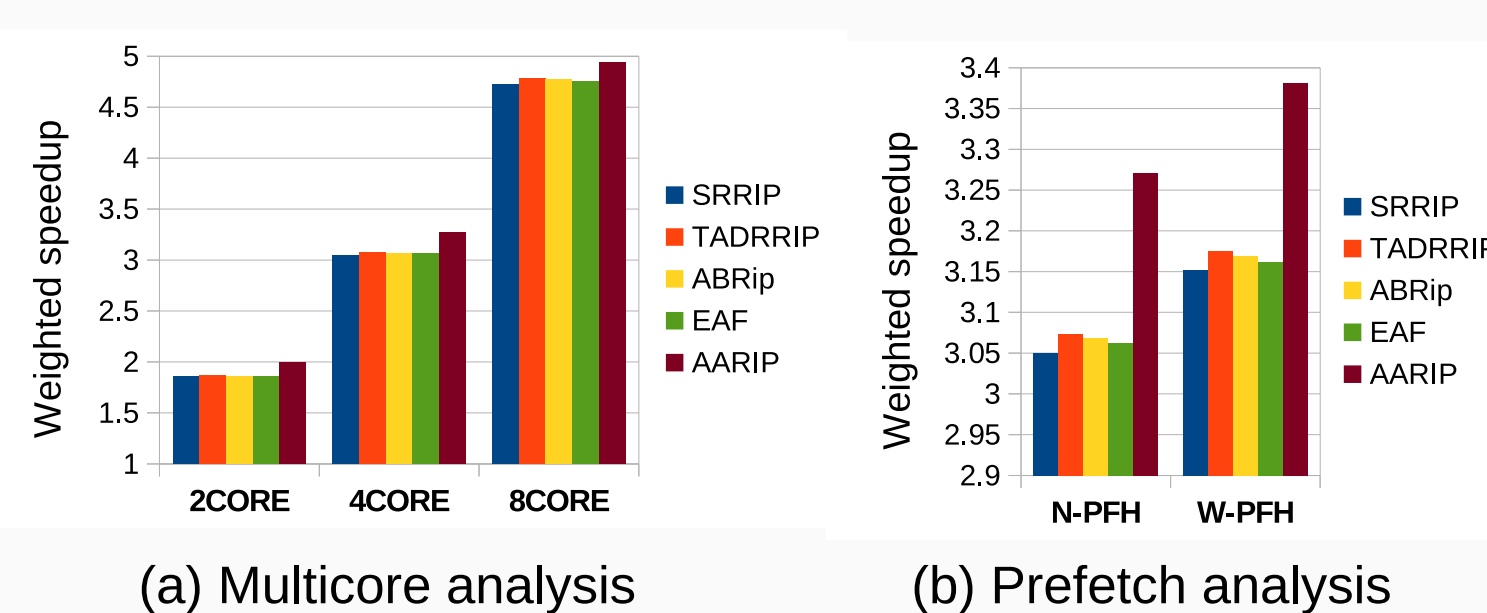
Fairness of shared resource allocation



Primary comparison: SRRIP / TADRRIP [1], ABRIP [2], EAF [3], and AARIP.

TAKEAWAY Application-aware victims + EbIS recovery preserve useful reuse.

MULTICORE + PREFETCH



Separate study: WS gains over SRRIP are **7.08% at 2 cores**, **4.48% at 8 cores**.

AARIP leads with next-line LLC prefetch **off (N-PFH)** or **on (W-PFH)**.

LOW STORAGE OVERHEAD

0.46% 4-core AARIP overhead

Additional storage over SRRIP; 4 MB, 16-way LLC.

	2-core	4-core	8-core
AARIP	0.44%	0.46%	0.48%
SHiP	0.68%	0.98%	1.56%
MRP	1.29%	2.16%	3.85%
Hawkeye	1.55%	2.52%	4.45%

Assumes existing coherence ownership bits. Versus these three policies, AARIP has **$\approx 2\%$ lower performance**, with much less storage.